BÁO CÁO THỰC HÀNH TUẦN 4 – OOP

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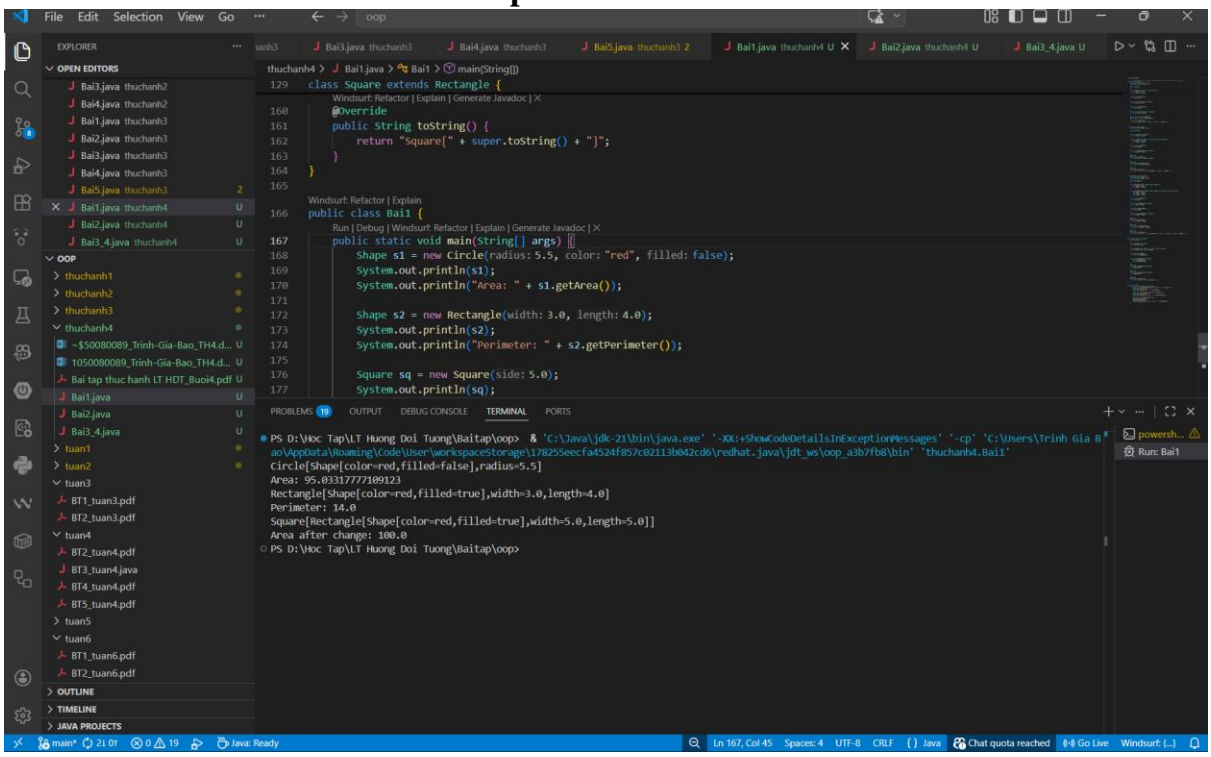
Bài 1: Kế thừa và Đa hình với Shape



Code:

```
package thuchanh4;


// 1. Xây dựng chương trình java gồm các lớp Shape, Circle, Rectangle, Square và kiểm tra

abstract class Shape {

    protected String color = "red";

    protected boolean filled = true;


    public Shape() {

    }
```

```
    public Shape(String color, boolean filled) {
        this.color = color;
        this.filled = filled;
    }

    public String getColor() {
        return color;
    }

    public void setColor(String color) {
        this.color = color;
    }

    public boolean isFilled() {
        return filled;
    }

    public void setFilled(boolean filled) {
        this.filled = filled;
    }

    public abstract double getArea();
    public abstract double getPerimeter();

    public String toString() {
        return "Shape[color=" + color + ",filled=" + filled + "]";
    }
}
```

```
class Circle extends Shape {
    protected double radius = 1.0;

    public Circle() {
    }

    public Circle(double radius) {
        this.radius = radius;
    }

    public Circle(double radius, String color, boolean filled) {
        super(color, filled);
        this.radius = radius;
    }

    public double getRadius() {
        return radius;
    }

    public void setRadius(double radius) {
        this.radius = radius;
    }

    @Override
    public double getArea() {
        return Math.PI * radius * radius;
    }
```

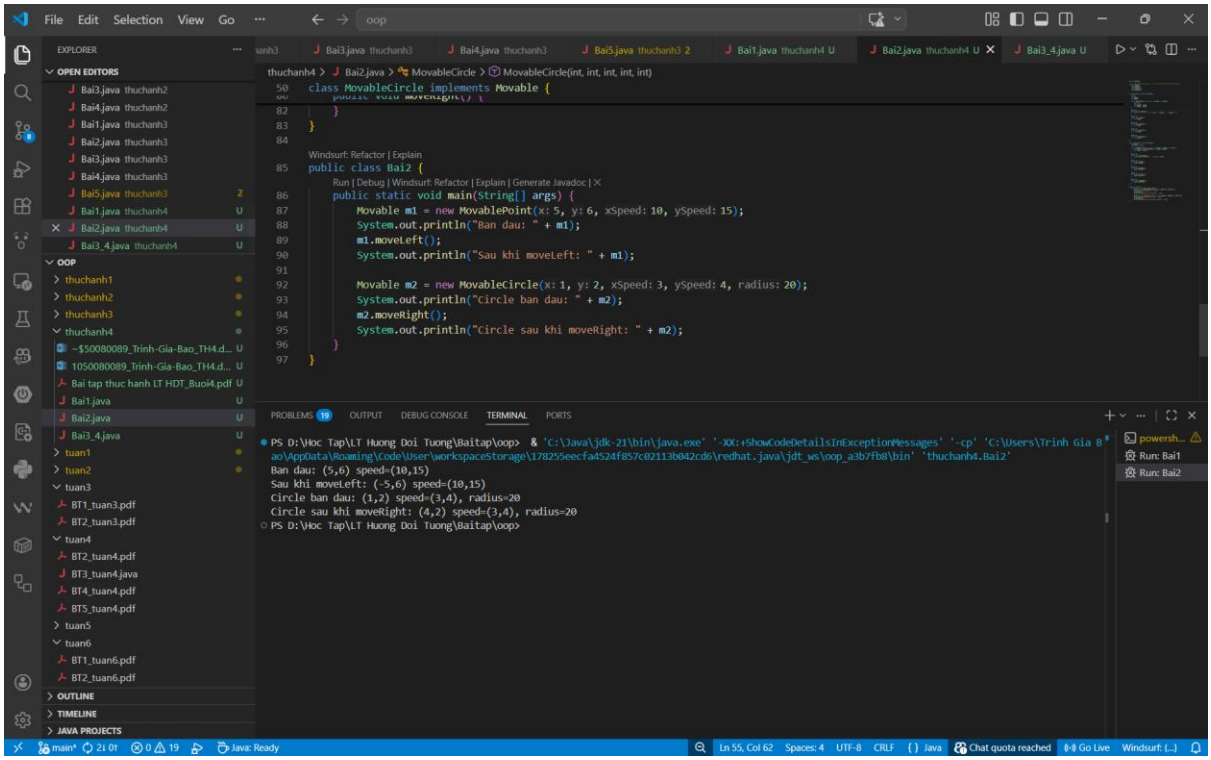
```
    @Override
    public double getPerimeter() {
        return 2 * Math.PI * radius;
    }

    @Override
    public String toString() {
        return "Circle[" + super.toString() + ",radius=" + radius + "]";
    }
}

class Rectangle extends Shape {
    protected double width = 1.0;
    protected double length = 1.0;

    public Rectangle() {
    }

    public Rectangle(double width, double length) {
        this.width = width;
        this.length = length;
    }

    public Rectangle(double width, double length, String color, boolean filled) {
        super(color, filled);
        this.width = width;
        this.length = length;
    }
```

```
public double getWidth() {
    return width;
}

public void setWidth(double width) {
    this.width = width;
}

public double getLength() {
    return length;
}

public void setLength(double length) {
    this.length = length;
}

@Override
public double getArea() {
    return width * length;
}

@Override
public double getPerimeter() {
    return 2 * (width + length);
}

@Override
public String toString() {
```

```
        return "Rectangle[" + super.toString() + ",width=" + width + ",length=" + length
+ "]";
    }
}

class Square extends Rectangle {
    public Square() {
    }

    public Square(double side) {
        super(side, side);
    }

    public Square(double side, String color, boolean filled) {
        super(side, side, color, filled);
    }

    public double getSide() {
        return super.getWidth();
    }

    public void setSide(double side) {
        super.setWidth(side);
        super.setLength(side);
    }

    @Override
    public void setWidth(double side) {
        setSide(side);
```

```
    }

    @Override
    public void setLength(double side) {
        setSide(side);
    }

    @Override
    public String toString() {
        return "Square[" + super.toString() + "]";
    }
}

public class Bai1 {
    public static void main(String[] args) {
        Shape s1 = new Circle(5.5, "red", false);
        System.out.println(s1);
        System.out.println("Area: " + s1.getArea());

        Shape s2 = new Rectangle(3.0, 4.0);
        System.out.println(s2);
        System.out.println("Perimeter: " + s2.getPerimeter());

        Square sq = new Square(5.0);
        System.out.println(sq);
        sq.setSide(10.0);
        System.out.println("Area after change: " + sq.getArea());
    }
}
```

Bài 2: Interface Movable



Code:

```
package thuchanh4;


// 2. Xây dựng Interface Movable và các lớp triển khai MovablePoint, MovableCircle
interface Movable {
    void moveUp();
    void moveDown();
    void moveLeft();
    void moveRight();
}


class MovablePoint implements Movable {
    int x;
    int y;
    int xSpeed;
    int ySpeed;
```

```
public MovablePoint(int x, int y, int xSpeed, int ySpeed) {
    this.x = x;
    this.y = y;
    this.xSpeed = xSpeed;
    this.ySpeed = ySpeed;
}

@Override
public String toString() {
    return "(" + x + "," + y + ") speed=(" + xSpeed + "," + ySpeed + ")";
}

@Override
public void moveUp() {
    y -= ySpeed;
}

@Override
public void moveDown() {
    y += ySpeed;
}

@Override
public void moveLeft() {
    x -= xSpeed;
}

@Override
```

```
    public void moveRight() {
        x += xSpeed;
    }
}

class MovableCircle implements Movable {
    int radius;
    MovablePoint center;

    public MovableCircle(int x, int y, int xSpeed, int ySpeed, int radius) {
        this.center = new MovablePoint(x, y, xSpeed, ySpeed);
        this.radius = radius;
    }

    @Override
    public String toString() {
        return center.toString() + ", radius=" + radius;
    }

    @Override
    public void moveUp() {
        center.moveUp();
    }

    @Override
    public void moveDown() {
        center.moveDown();
    }
```

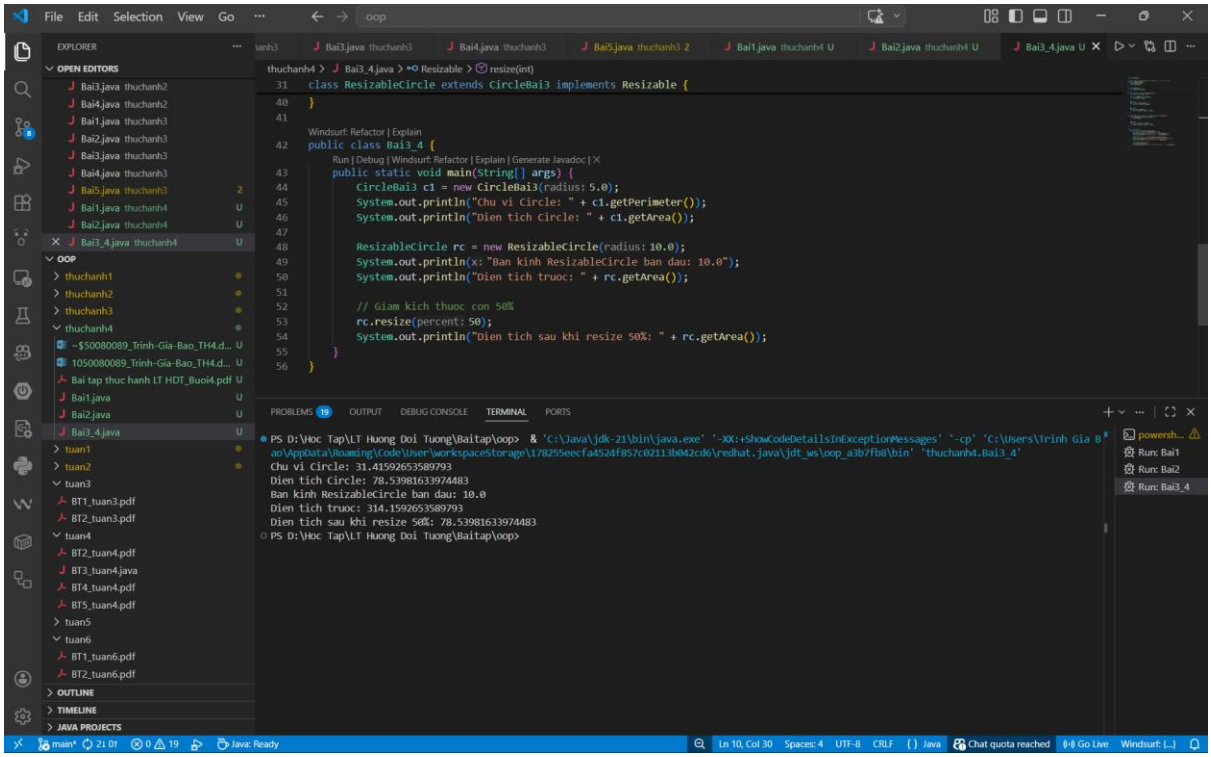
```
    @Override
    public void moveLeft() {
        center.moveLeft();
    }

    @Override
    public void moveRight() {
        center.moveRight();
    }
}

public class Bai2 {
    public static void main(String[] args) {
        Movable m1 = new MovablePoint(5, 6, 10, 15);
        System.out.println("Ban dau: " + m1);
        m1.moveLeft();
        System.out.println("Sau khi moveLeft: " + m1);

        Movable m2 = new MovableCircle(1, 2, 3, 4, 20);
        System.out.println("Circle ban dau: " + m2);
        m2.moveRight();
        System.out.println("Circle sau khi moveRight: " + m2);
    }
}
```

Bài 3 & 4: Interface GeometricObject và Resizable

Lý do em gộp chung Bài 3 và Bài 4 vào một cụm là vì:

Class ResizableCircle ở Bài 4 bắt buộc phải kế thừa gốc gác từ class Circle của Bài 3 thì hệ thống mới trơn tru được. Tách riêng tụi nó ra là IDE "khóc thét" báo lỗi không tìm thấy class cha ngay. Nên em cho tụi nó về chung một nhà luôn để code vừa gọn

gàng, logic lại phát huy triệt để tính kế thừa và interface. Gom lại thế này em chỉ cần viết một hàm main chạy test nghiệm thu 1 lần là ra ngay kết quả, bao mượt bao slay luôn ạ.



Code:

```
package thuchanh4;

// 3 & 4. Interface GeometricObject, Resizable và các lớp triển khai
interface GeometricObject {
    double getPerimeter();
    double getArea();
}

interface Resizable {
    void resize(int percent);
}

class CircleBai3 implements GeometricObject {
```

```
    protected double radius = 1.0;

    public CircleBai3(double radius) {
        this.radius = radius;
    }

    @Override
    public double getPerimeter() {
        return 2 * Math.PI * radius;
    }

    @Override
    public double getArea() {
        return Math.PI * radius * radius;
    }
}

class ResizableCircle extends CircleBai3 implements Resizable {
    public ResizableCircle(double radius) {
        super(radius);
    }

    @Override
    public void resize(int percent) {
        radius = radius * percent / 100.0;
    }
}

public class Bai3_4 {
```

```
    public static void main(String[] args) {
        CircleBai3 c1 = new CircleBai3(5.0);
        System.out.println("Chu vi Circle: " + c1.getPerimeter());
        System.out.println("Dien tich Circle: " + c1.getArea());

        ResizableCircle rc = new ResizableCircle(10.0);
        System.out.println("Ban kinh ResizableCircle ban dau: 10.0");
        System.out.println("Dien tich truoc: " + rc.getArea());

        // Giam kich thuoc con 50%
        rc.resize(50);
        System.out.println("Dien tich sau khi resize 50%: " + rc.getArea());
    }
}
```